

1 Range Processing

Definition 1.1 (Bandwidth). *Radar bandwidth (frequency) B available to be swept during the chirp period; typically MHz or GHz.*

Remark 1.1 (TI Implementation). On a TI radar, this is `freq_slope * adc_samples / sample_rate`.

Definition 1.2 (Speed of Light). $c = 299792458 \text{ m/s}$.

Definition 1.3 (Sample rate). *Sampling frequency F_s of the radar ADC.*

Remark 1.2 (Units). This is typically specified in KHz (Ksps) or MHz (Msps).

Definition 1.4 (Number of samples). *Number of ADC samples N per chirp; this leads to a total observation time of $T_s = N/F_s$.*

Definition 1.5 (Slope). *Slope $S = B/T_s = BF_s/N$ of the chirp waveform; typically MHz/ μs .*

Definition 1.6 (Range Resolution). $\Delta r = \frac{c}{2B}$.

Suppose a given object is at range r . This leads to time delay $\tau = 2r/c$, so $\Delta r = c\Delta\tau/2$.

When taking a DFT over the chirp, the frequency resolution is $1/T_s$. Since the frequency resolution corresponding to a time resolution is S , the time delay resolution is $\Delta\tau = \frac{1/T_s}{S} = \frac{1}{B}$. This leads to $\Delta r = c\Delta\tau/2 = \frac{c}{2B}$.

Definition 1.7 (Maximum Range). $R_{\max} = N \frac{c}{2B} = \frac{F_s c}{2S}$.

This can be seen by combining the range resolution with the number of range bins, which is equal to the DFT window size (N).

Remark 1.3 (Padding). Zero-padding the N ADC samples to $N' > N$ before the range FFT increases the number of range bins via spectral interpolation, but does not change the true range resolution $\Delta r = c/2B$ (set by the swept bandwidth B) or the true maximum unambiguous range (set by the sample rate F_s and slope S).

Remark 1.4 (Cropping). Using fewer than N of the acquired samples shortens the effective bandwidth B and observation window T_s , coarsening the range resolution Δr proportionally (since the frequency resolution of the shorter DFT is $1/T'_s$ for the cropped window $T'_s < T_s$), and reduces the number of range bins accordingly.

Remark 1.5 (Decimation). Keeping every D -th ADC sample reduces the effective sample rate to F_s/D ; the observation window T_s is unaffected, so the range resolution $\Delta r = c/2B$ is unchanged, but the maximum unambiguous range $R_{\max} = F_s c/(2S)$ shrinks by a factor of D , risking range aliasing unless the signal is properly low-pass filtered first.

2 Doppler Processing

Definition 2.1 (Wavelength). $\lambda = c/F$.

Remark 2.1 (Narrow-Band Approximation). If using a narrow-band radar, λ can be approximated as a constant with respect to the center frequency of the chirp.

Definition 2.2 (Chirp Rate). *Time between chirps T_c .*

Remark 2.2 (TI Implementation). On a TI radar, if using time-division multiplexing across antennas, this is `(idle_time + ramp_end_time)* num_tx`.

Definition 2.3 (Doppler Resolution). $\Delta v = \frac{\lambda}{2N_d T_c}$.

Suppose a given object has radial velocity v . The two-way Doppler shift is $f_d = 2v/\lambda$, so $v = f_d\lambda/2$ and $\Delta v = \Delta f_d\lambda/2$.

When taking a DFT over the N_d chirps of a frame (slow time), the frequency resolution is $1/(N_d T_c)$. This is exactly the Doppler frequency resolution Δf_d , so $\Delta v = \Delta f_d\lambda/2 = \frac{\lambda}{2N_d T_c}$.

Definition 2.4 (Maximum Doppler Velocity). $v_{\max} = \frac{\lambda}{4T_c}$.

Chirps are sampled once every T_c (slow-time sampling rate $1/T_c$), so by the Nyquist criterion the unambiguous Doppler frequency is limited to $|f_d| \leq \frac{1}{2T_c}$. Since $v = f_d\lambda/2$, this gives $v_{\max} = \frac{1}{2T_c} \cdot \frac{\lambda}{2} = \frac{\lambda}{4T_c}$.

Remark 2.3 (Padding). Zero-padding the N_d chirps to $N'_d > N_d$ before the Doppler FFT increases the number of Doppler bins via spectral interpolation, but does not change the true Doppler resolution $\Delta v = \lambda/(2N_d T_c)$ (set by the number of chirps actually collected) or the maximum unambiguous velocity.

Remark 2.4 (Cropping). Using fewer than N_d of the collected chirps shortens the coherent processing interval $N_d T_c$, coarsening the Doppler resolution $\Delta v = \lambda/(2N_d T_c)$ proportionally, and reduces the number of Doppler bins accordingly.

Remark 2.5 (Decimation). Keeping every D -th chirp increases the effective chirp period to DT_c ; since the total coherent processing interval $N_d T_c$ is unchanged, the Doppler resolution $\Delta v = \lambda/(2N_d T_c)$ is preserved, but the maximum unambiguous velocity $v_{\max} = \lambda/(4T_c)$ shrinks by a factor of D , risking Doppler aliasing unless the target's true velocity is known to lie within the new (narrower) unambiguous range.

3 Range-Doppler Geometry

3.1 Inputs

Definition 3.1 (Velocity). We denote the velocity direction as $\mathbf{v}$, where $\|\mathbf{v}\|_2 = 1$.

Definition 3.2 (Speed). Denote the magnitude component as s , where the full velocity is $\mathbf{v}s$.

Definition 3.3 (Measurements: Distance (radius), Doppler). (r, d) bins; assuming the sensor position is $\mathbf{0}$, for positions $\mathbf{a}$, this constrains measurements to $\|\mathbf{a}\|_2 = r$ and $\frac{\mathbf{a}^T \mathbf{v}s}{\|\mathbf{a}\|_2} = \frac{\mathbf{a}^T \mathbf{v}s}{r} = d$.

The target integration is described by

$$y(r, d; \mathbf{x}_{\text{sensor}}, \mathbf{v}) = \int_{\|\mathbf{t}\|_2=r, \frac{\mathbf{t}^T \mathbf{v}}{\|\mathbf{t}\|_2}=d} \sigma(\mathbf{x}_{\text{sensor}} + \mathbf{t}) d\mathbf{t}$$

3.2 Integration Circle

The constraints are a sphere and a plane, which should constrain measurements to a circle. The center of the circle is $\mathbf{c} = \mathbf{v} \frac{rd}{s}$, since this is in the direction of the normal vector $\mathbf{v}$ and satisfies

$$\frac{(\mathbf{v}(rd/s))^T \mathbf{v}s}{r} = \frac{\mathbf{v}^T \mathbf{v}s(rd/s)}{r} = d.$$

The radius of this circle is

$$r' = \sqrt{r - \left\| \mathbf{v} \frac{rd}{s} \right\|^2} = \sqrt{r - \left(\frac{rd}{s} \right)^2} = r \sqrt{1 - \left(\frac{d}{s} \right)^2}.$$

Let $\mathbf{p}$ and $\mathbf{q}$ form an orthonormal basis with $\mathbf{v}$ (i.e. via Gram-Schmidt). Then, the circle can be parameterized by angle ψ as

$$\begin{aligned} \mathbf{a} &= r' \mathbf{p} \cos(\psi) + r' \mathbf{q} \sin(\psi) + \mathbf{v} \frac{rd}{s} \\ &= r \left(\sqrt{1 - (d/s)^2} (\mathbf{p} \cos(\psi) + \mathbf{q} \sin(\psi)) + \mathbf{v} \frac{d}{s} \right). \end{aligned}$$

4 Angular FFT Processing

Definition 4.1 (Antenna Spacing). *Spacing d between elements of the (virtual) receive array.*

Remark 4.1 (Typical Value). Radars typically set to $d = \lambda/2$ to avoid spatial aliasing.

Definition 4.2 (Number of Antennas). *Number of (virtual) receive antenna elements N_a in the array.*

Let θ be the angle to a target, measured from broadside (i.e., the array normal) of a linear array with N_a elements and spacing d . After range-Doppler processing, a given range-Doppler bin contains one complex sample per receive antenna; the extra path length to antenna n relative to antenna 0 is $nd \sin \theta$, giving a phase offset of $\frac{2\pi}{\lambda} nd \sin \theta$. The spatial signal across antennas is therefore

$$x[n] = A \exp \left(j \frac{2\pi}{\lambda} nd \sin \theta \right), \quad n = 0, \dots, N_a - 1.$$

Taking an N_a -point DFT over this spatial signal (i.e., across antennas for a fixed range-Doppler bin) yields the angular spectrum

$$X[k] = \sum_{n=0}^{N_a-1} x[n] \exp \left(-j \frac{2\pi}{N_a} kn \right),$$

which is maximized (constructive interference) when $\frac{k}{N_a} = \frac{d \sin \theta}{\lambda}$.

Definition 4.3 (Angular Bin Mapping). *Bin k of the angular spectrum corresponds to $\sin \theta = \frac{k\lambda}{N_a d}$; with the typical spacing $d = \lambda/2$, this is $\sin \theta = \frac{2k}{N_a}$.*

Remark 4.2 (Non-linearity). Unlike range and Doppler, the mapping from bin index to angle is non-linear in θ (linear only in $\sin \theta$), so angular bins are spaced increasingly far apart in θ towards $\pm 90^\circ$.

Remark 4.3 (2D Arrays). For 2D arrays, the same principle applies in both dimensions, with the angular spectrum being a 2D DFT over the 2D array, and the angles corresponding to each bin being determined by the number of antennas and spacing in each dimension.

Definition 4.4 (Angular Resolution). $\Delta(\sin \theta) = \frac{\lambda}{N_a d}$.

As with Doppler, the DFT over the N_a -antenna aperture has a (normalized spatial) frequency resolution of $1/N_a$ in $\frac{d \sin \theta}{\lambda}$, so $\Delta \left(\frac{d \sin \theta}{\lambda} \right) = \frac{1}{N_a}$, giving $\Delta(\sin \theta) = \frac{\lambda}{N_a d}$.

This resolution is set by the physical aperture $N_a d$: two targets closer than $\Delta(\sin \theta)$ apart cannot be separated into distinct peaks.

Remark 4.4 (Padding). Zero-padding the N_a antenna samples to $N'_a > N_a$ before the angular FFT increases the number of angular bins via spectral interpolation, but does not change the true angular resolution $\Delta(\sin \theta) = \lambda/(N_a d)$ (set by the physical aperture $N_a d$).

Remark 4.5 (Cropping). Using fewer than N_a of the array elements shortens the effective aperture, coarsening the angular resolution $\Delta(\sin \theta)$ proportionally, and reduces the number of angular bins accordingly.

Remark 4.6 (Decimation). Keeping every D -th antenna element over the same physical aperture increases the effective spacing to Dd ; since the aperture $N_a d$ is unchanged, the angular resolution $\Delta(\sin \theta) = \lambda/(N_a d)$ is preserved, but exceeding the $d = \lambda/2$ spatial Nyquist limit introduces grating lobes (spatial aliasing), causing ambiguous angle estimates.